

CS101

For-Loops

2/26

Agenda

1. Quiz5
2. Demo c/python/java
3. Demo healthPrinter (if/else)
4. Demo for-loops
5. Lab7 Work time
6. Windows OS Tips + Touch Typing

Quiz5

CS101 Trajectory

- Cover all basic concepts first
- Still need
 - While loops
 - Arrays
 - Methods/Functions
- Then practice with more interesting problems
- Continue to sprinkle in tangential topics

Demo

java/C/python

STATICALLY TYPED • DYNAMICALLY TYPED

CATEGORY	• STATICALLY TYPED	• DYNAMICALLY TYPED
DEFINITION	Types are known at compile time. Once declared, a variable's type cannot change.	Types are determined at runtime. A variable can hold different types throughout its lifetime.
LANGUAGES	Java C C++ Go Rust	Python JavaScript Ruby PHP
CODE EXAMPLE	JAVA <pre>int x = 5; x = "hello"; // compile error</pre>	PYTHON <pre>x = 5 x = "hello" # no error</pre>
PROS	+ Catches type errors before running + Better performance (optimized at compile time) + Easier to maintain large codebases	+ Faster to write – less boilerplate + More flexible and forgiving + Great for prototyping and scripting
CONS	- More verbose – must declare types - Slower to write for small scripts	- Type errors only found at runtime - Slower execution speed - Harder to debug in large projects

Demo

healthPrinter.java

Demo for-loops

Youtube Tips

ShortCut Keys Popup in Settings

- **Left/Right Arrow** → Forward/Back
- **Spacebar** → Pause
- **F** → Toggle fullscreen
- **>** → Speed Up
- **<** → Slow down

General → Hold shift and scroll to go left/right

Lab7 Work time

HW3

Mario Blocks

Announcements

1. Submit HW2 and Lab3!
2. Lab4 due Friday 2/27 at midnight
3. Bring completed Lab6 on Tuesday